

7(a)(i)	an electric current that periodically reverses its direction in a circuit (with a frequency)	B1
7(a)(ii)	- wire carries a current that is perpendicular to the magnetic field between the magnet poles - (by Fleming's LHR) wire experiences a force that is normal to both the current and magnetic field - force experienced is directed up and down vertically - since current reverses direction periodically, vibration is the same frequency as the AC	B1 B1 B1 B1
7(a)(iii)1.	recognising $\lambda = 2(L_{XP})$ $v = f\lambda = (120)(4) = 480 \text{ m s}^{-1}$	M1 A1
7(a) (iii)2.	- <u>meeting</u> of (waves) - waves of same type, same frequency, opposite direction (characteristics) - incident waves from force on AC-carrying wire in magnetic field reflected off X and P (Origin of standing waves) - speed of incident / reflected wave in wire (concluding with correct answer)	B1 B1 B1 B1

Question	Marks
7(a) (iv) $v = f\lambda$, therefore $v \propto f$ because λ is fixed by length of XP tension is provided by weight, $T = mg$ therefore $T \propto m$	B1 B1
Need to show that $f \propto \sqrt{m}$ Take ratio:	
$\frac{f_1}{f_0} = \frac{170}{120} \approx 1.42, \sqrt{\frac{2m}{m}} = \sqrt{2} \approx 1.41$	M1
$\frac{f_2}{f_0} = \frac{208}{120} \approx 1.73, \sqrt{\frac{3m}{m}} = \sqrt{3} \approx 1.73$	
Likely valid	A0
7(b)(i) waves spread at edge or slit changes direction and spreads into geometric shadow	B1
7(b)(ii) $\theta = \frac{1}{2}(188 - 160) = 14^\circ$	C1
$d \sin \theta = n\lambda \rightarrow d \sin(14^\circ) = 2(633 \times 10^{-9})$	M1
$d = 5.23 \times 10^{-6} \text{ m}$	C1
$d = \frac{1}{N} \rightarrow N = \frac{1}{d} = 1.91 \times 10^5$	A1
7(b) (iii) each bright fringe is constructive interference from multiple points of diffracted waves (vs two points in double slit)	B1